# Practice Questions

## MTH633

## Lectures ( 32 – 34 ) or Modules ( 127 – 139 )

**Q1:** Compute the factor group $(Z_4 \times Z_6) / \langle (0, 1) \rangle$.

**Q2:** Show that the group $G = Z / 12Z$ is not simple.

**Q3:** Define commutator subgroup.

**Q4:** Let $G$ be a group. The set of all commutators $aba^{-1}b^{-1}$ for $a, b \in G$ generates a subgroup $C$ of $G$.